Inkloom AI Challenge – Round 1 Submission (“The Plan”)

BrandGuard AI

Intelligent Pre-Publication Brand Governance Platform

Team Xnihilo

Team Members

Name	Role
Subhajit Das	Team Leader – Multimodal RAG & AI Architecture
Sambu Chowdhury	Core AI Engineer – Branched RAG & Static Audit Pipeline
Rana Tarafdar	Full-Stack Engineer – Frontend & FastAPI Backend Bridge
Mousumi Bag	UI/UX Engineer – Frontend Review Portal & Approval Workflow

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1. PROBLEM DEFINITION & EVIDENCE THE PROBLEM IS REAL

1.1 The Core Problem

Growing organisations publish content at high velocity through internal employees and external marketing agencies. Because brand rules live in static, disconnected PDF files and spreadsheets, content frequently goes live without unified verification. This creates brand inconsistency and credibility damage across **six critical dimensions**:

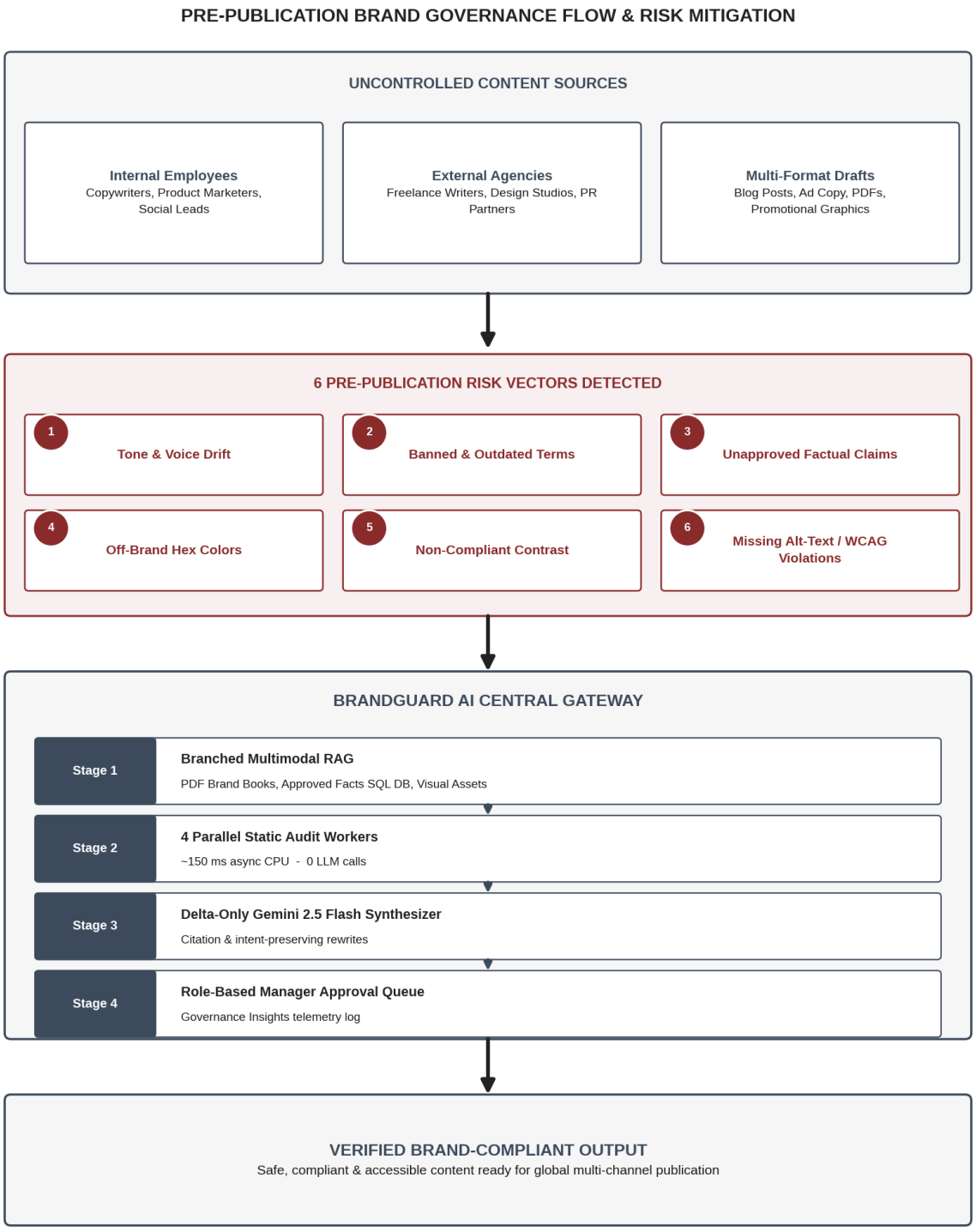
#	Dimension	What goes wrong
1	Inconsistent Tone & Brand Voice	Copy written by different employees or agencies drifts away from the organisation's approved persona.
2	Banned & Inconsistent Terminology	Use of deprecated product names, unapproved industry jargon, or restricted promotional buzzwords (e.g., "cheap", "guaranteed").
3	Outdated Factual Information	Publishing legacy pricing tiers, retired feature specifications, or outdated company metrics.
4	Unsupported Promotional Claims	Making unverifiable superlative claims (e.g., "#1 fastest platform", "99.9% cost reduction") without approved source backing.
5	Off-Brand Visual Consistency	Marketing banners and promotional graphics using unapproved Hex color codes or stretched logos.
6	Accessibility Non-Compliance	Publishing visuals and documents that fail WCAG 2.1 AA/AAA color-contrast standards (minimum 4.5:1) or omit alt-text.

1.2 Visual Illustration 1: The Brand Governance Gateway & Pre-Publication Flow

The table below maps each content source to the risk vectors it creates and the gateway stage that addresses them; the illustration that follows shows the same flow visually.

Uncontrolled Content Source	Pre-Publication Risk Vector	BrandGuard AI Central Gateway
Internal Employees (Copywriters, Product Marketers, Social Media Leads)	1. Tone & Voice Drift: inconsistent brand personality across channels	Stage 1 – Branched Multimodal RAG: indexes Brand Voice PDFs, Approved Facts SQL, and Visual Asset Rules
External Agencies (Freelance Writers, Design Studios, PR Partners)	2. Terminology Violations: banned words, slang, and deprecated product names	Stage 2 – Static Worker 1 (Lexical & Style): sub-150ms Regex/Aho-Corasick banned-term scan + Textstat tone metrics
Content Management & Publishing Teams	3. Outdated Facts: legacy pricing, old metrics, and retired feature specs	Stage 3 – Static Worker 2 (Fact Verification): automated claim extraction cross-checked against Approved Facts SQL DB
Marketing Graphic Creators & Designers	4. Unsupported Claims: unverified superlatives ("#1", "99% faster") causing legal risk	Stage 4 – Static Worker 3 (Visual Audit): OpenCV/Pillow K-Means Hex Delta-E palette check + EasyOCR banner scan
Distributed Publishing Workflows (Web Docs, CMS, PDFs)	5. Off-Brand Visuals: unapproved Hex color palettes and improper logo usage	Stage 5 – Static Worker 4 (Accessibility): deterministic WCAG 2.1 AA/AAA contrast math + alt-text & readability check
Direct Publishing Channels (Blogs, Press Releases, Socials)	6. Accessibility Failures: low contrast (<4.5:1), missing alt-text, complex readability	Stage 6 – Delta-Only Gemini Synthesis & Manager Approval: explains cited rules, suggests rewrites, and enforces sign-off

Illustration 1



Pre-publication brand governance flow and risk mitigation – from uncontrolled content sources to verified, compliant output.

2. TARGET USERS, THEIR NEEDS & ORGANISATIONAL SETTING

2.1 Organisational Setting

BrandGuard AI is designed as a centralized web platform for scaling startups, mid-sized enterprises, and digital marketing agencies where internal teams and external contributors produce multi-format content (blog posts, ad copy, PDF brochures, social graphics) within a structured review workflow.

2.2 Target User Personas & Specific Needs

Persona	Organisational Role	Core Pain Points	What They Need from BrandGuard AI
Persona A: Content Creators (Writers, Designers, Agency Contributors)	Drafts daily articles, ad copy, social posts, and promotional graphics in the web portal	Frustrated by 50-page brand PDFs, slow manager rejections, and vague feedback that forces full rewrites	Real-time, constructive explanations citing exact rules, plus 1-click rewrite suggestions that preserve their original creative purpose
Persona B: Authorised Reviewers (Brand Managers, Legal & Compliance Leads)	Reviews and approves content prior to publication to protect brand identity and legal compliance	Spends 4–6 hours/week manually checking hex codes, statistics, and banned terms across dozens of drafts	Pre-screened drafts with highlighted issues, side-by-side diff comparison, inline commenting, revision history, and role-based sign-off
Persona C: Brand Authorities (CMOs, Brand Directors, Non-Tech Admins)	Defines official brand guidelines, approved facts, terminology dictionaries, and visual asset kits	No visibility into which brand rules are repeatedly broken or where guidelines confuse creators	A zero-code drag-and-drop vault to upload guidelines/facts/assets and a Governance Insights Dashboard tracking recurring mistakes

3. PUBLIC SOCIAL MEDIA DISCUSSIONS & COMMUNITY EVIDENCE

An analysis of public social media posts and community discussions across Reddit (r/marketing, r/copywriting, r/graphic_design), LinkedIn marketing forums, and X/Twitter threads reveals widespread, real-world operational friction caused by static brand guidelines and manual approval bottlenecks:

1. Frustration with Static PDFs & Disrupted Creative Flow (Reddit & X/Twitter Discussions)

- Creators frequently post about ignoring 50+ page PDF style guides because searching through complex PDF documents while drafting content severely interrupts writing momentum.
- Marketing professionals on LinkedIn regularly advocate for centralized automated tools that pre-screen drafts and offer instant, line-by-line compliance checks before manager submission.

2. Reviewer Fatigue & Slow Review Cycles (LinkedIn & Marketing Forum Records)

- Brand managers report spending hours weekly manually flagging repetitive errors, including deprecated terminology, unverified superlatives, off-brand color Hex codes, and low-contrast text overlays.
- Discussions emphasize that outright rejections create friction; reviewers need systems that explain rule violations and provide constructive, compliant rewrite alternatives.

3. Widespread Brand Drift in Published Digital Content (Search & Social Media Benchmarks)

- Public content audits discussed across design and marketing forums show high rates of unverified promotional claims and promotional graphics failing WCAG 2.1 AA contrast standards (4.5:1) when published without automated verification.

4. PROPOSED SOLUTION, VALUE PROPOSITION & END-TO-END WORKFLOW

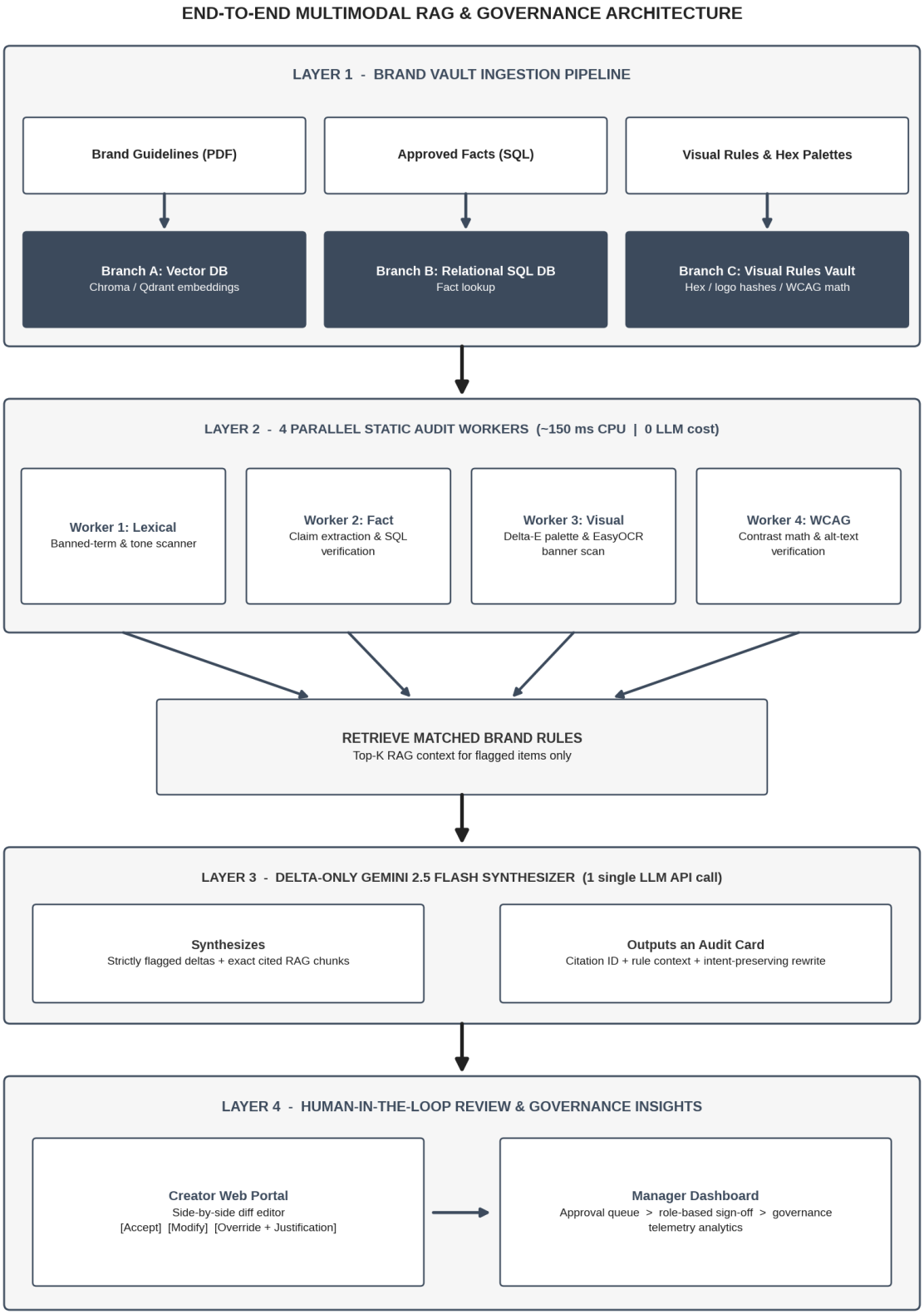
4.1 Core Value Proposition

BrandGuard AI is an intelligent, centralized web platform consisting of:

1. **A Central Multimodal Brand Vault & Collaborative Web Portal:** where non-technical Brand Authorities upload guidelines, approved facts, terminology, and visual assets, and where Creators submit drafts (text/images) for automated pre-screening.
2. **A 4-Worker Deterministic Pre-Check + Branched RAG + Delta-Only LLM Engine:** runs 4 parallel static Python workers in ~150ms (0 LLM calls) to detect exact violations, queries a Branched Multimodal RAG for the matching brand rules, and sends only the flagged snippets ("Delta-Only") to 1 single Google Gemini call to generate clear explanations and intent-preserving rewrites.
3. **A Role-Based Manager Approval & Revision Workflow:** a clean dashboard where managers collaborate through side-by-side diffs, comments, and approvals, backed by a Governance Insights telemetry log.

4.2 Visual Illustration 2: End-to-End Upgraded Multimodal RAG & Governance Architecture

Illustration 2



End-to-end multimodal RAG and governance architecture – four layers from Brand Vault ingestion to human-in-the-loop review.

5. PLANNED FEATURES, TECHNOLOGY STACK, INTEGRATIONS & DATA SOURCES

Processing Pipeline Overview

Supplementary diagram



The four processing stages: Brand Vault ingestion, parallel static audit, delta-only Gemini synthesis, and human-in-the-loop approval.

5.1 Complete Feature Matrix Mapped to Suggested Product Capabilities

Inkloom AI Required Capability	BrandGuard AI (Team Xnihilo) Planned Implementation
Import or define brand guidelines, approved facts, terminology & reusable assets	Drag-and-drop Brand Vault supporting PDF style guides, structured Approved Facts tables, Banned/Preferred Term dictionaries, Hex color codes, and official logo files.
Review text, images, documents or other supported content formats	Centralized web upload gateway accepting rich text/Markdown, multi-page PDF documents, and PNG/JPEG promotional graphics (including OCR text inside images).
Detect inconsistent tone, language, facts, claims, visual usage & accessibility	4 Parallel Static Workers + Branched RAG covering all 6 dimensions simultaneously in under 3 seconds end-to-end.
Explain why an issue was flagged and identify the guideline or source involved	Every flag displays a structured Inspector Card citing the exact PDF document name, page number, section heading, or Fact ID (e.g., "Brand Book v2, Page 7, Section 4.2").
Suggest corrections without automatically overriding human decisions	Non-destructive side-by-side diff suggestions; creators explicitly click [Accept], [Edit], or [Override with Comment] – the AI never overwrites content silently.
Support comments, approvals, revision history & role-based workflows	Role-based workspaces (Creator, Reviewer, Manager) with inline comment threads, version snapshots (v1, v2, v3), and status transitions (Draft → In Review → Changes Requested → Approved).
Identify recurring problems and provide useful governance insights over time	Governance Insights Analytics Dashboard visualizing top violated rules, recurring terminology mistakes, and "Ambiguity Hotspots" where creators frequently override a rule.
Protect confidential brand materials & disclose AI/external services	Delta-Only snippet transmission, 1-click Local/Offline LLM mode (Ollama), and explicit AI attribution badges on every generated review card.

5.2 The Three Named Improvements (Tracked Across Rounds 2 & 3)

To directly align with the Round 2 (15 pts) and Round 3 (20 pts) judging rubrics, Team Xnihilo defines three high-impact improvements upfront:

1. **Improvement 1 – Centralized Gateway Web Portal:** a clean, lightning-fast web app where creators drop documents/images for instant RAG audit without needing complex browser extensions.
2. **Improvement 2 – Deterministic Visual & WCAG Accessibility Engine:** combines OpenCV/Pillow K-Means color clustering (ΔE color distance against approved Hex palettes), perceptual logo hashing, EasyOCR banner text scanning, and mathematical WCAG 2.1 AA/AAA contrast verification.
3. **Improvement 3 – Confidential Local-Edge Privacy Mode & Governance Ambiguity Detector:** lets organisations toggle from Cloud Gemini APIs to 100% local execution (Ollama / WebLLM) for sensitive pre-release materials, while automatically flagging brand guidelines that trigger high creator override rates as "Unclear Guidelines Needs Revision."

5.3 100% Free-Tier ($0 Cost) Technology Stack

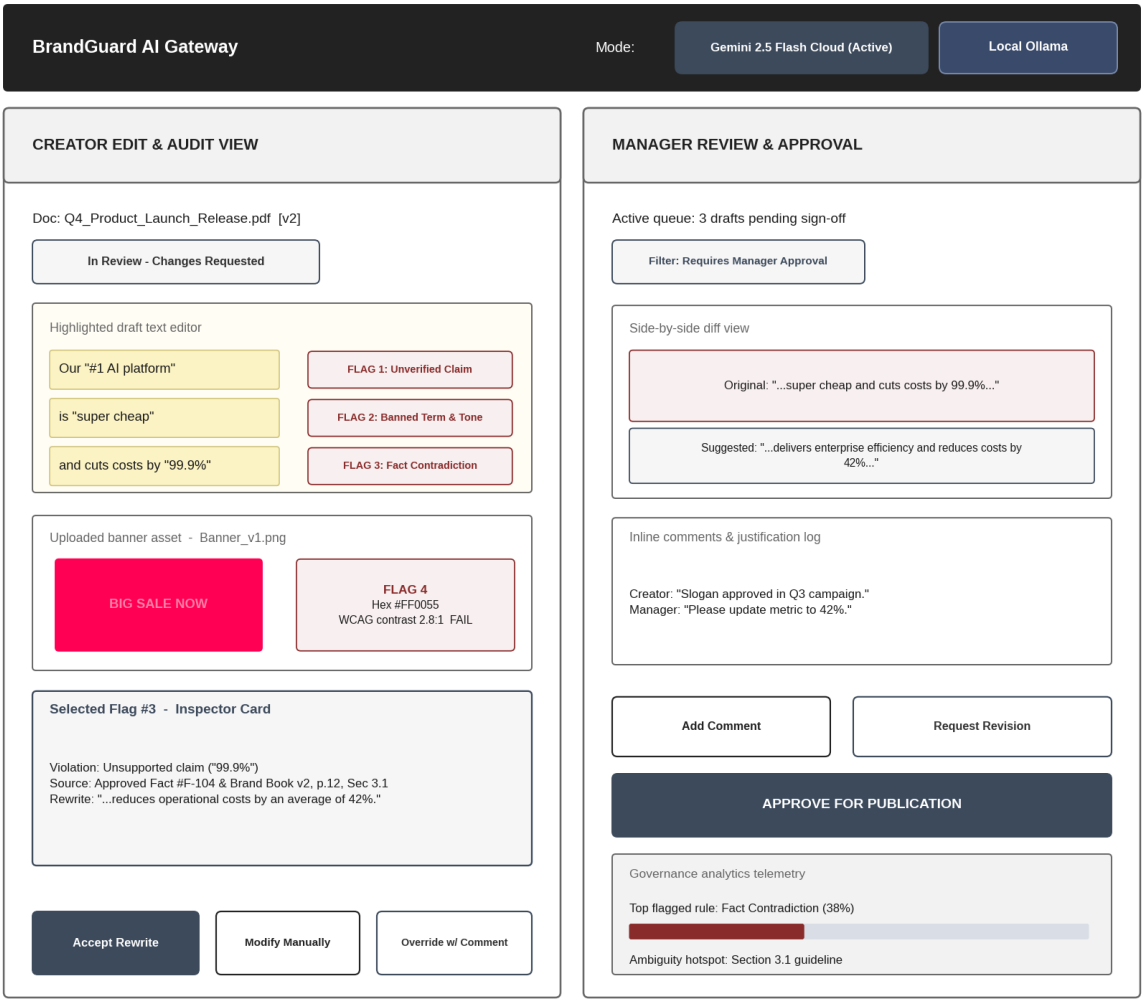
Architecture Layer	Technologies & Libraries	Purpose & Technical Justification
Frontend Web Portal	Next.js (React), TypeScript, Tailwind CSS, Shadcn UI, Recharts (hosted on Vercel Free Tier)	Delivers a responsive side-by-side diff editor, Manager approval queue, version history viewer, and Governance Insights analytics charts.
Backend API Server	Python 3.11, FastAPI, asyncio (hosted on HuggingFace Spaces Docker / Render Free Tier)	Orchestrates parallel static workers, exposes REST endpoints (/upload-guidelines, /audit-content, /approve-draft, /analytics) for the web portal.
Branched RAG & Embeddings	ChromaDB / Qdrant Cloud (Free Tier), Google Gemini Text Embeddings / BAAI/bge-small-en-v1.5, PyMuPDF (fitz)	Extracts and indexes unstructured Brand Guidelines PDFs with exact page/section metadata for zero-hallucination citations.
4 Parallel Static Workers	Python re, Aho-Corasick, textstat, Pillow, OpenCV (cv2), imagehash, EasyOCR, wcag-contrast-ratio	Executes deterministic, zero-LLM-cost checks for banned terms, reading complexity, Hex palette Delta-E distance, logo usage, and WCAG contrast.
Delta-Only LLM Synthesis	Google Gemini 2.5 Flash / Pro API (Google AI Studio Free Tier) + Ollama (local confidential fallback)	Processes only flagged snippets + retrieved RAG rules to generate human-friendly explanations and intent-preserving rewrites.
Database, Auth & Audit Logs	Supabase (Free Tier PostgreSQL + Auth + Storage)	Stores Approved Facts tables, Role-Based Access Control (Creator/Reviewer/Manager), inline comments, revision history, and violation telemetry.
AI-Assisted Development Workflow	Google Gemini Models + Antigravity IDE (used strictly within the official build window with Human-in-the-Loop testing)	Accelerates boilerplate generation, UI scaffolding, and unit test creation while every team member manually tests and verifies all logic.

5.4 Data Sources & External Integrations

- **Primary Data Sources:** Organisation-uploaded Brand Guideline PDFs, Approved Factual Claims CSV/SQL tables, Banned & Preferred Terminology JSON dictionaries, Approved Brand Hex Palette definitions, and Official Logo/Asset packs (plus synthetic benchmark brand kits created by Team Xnihilo for judge testing).
- **Integrations:** (1) Google Gemini API for multimodal reasoning and embeddings; (2) Supabase PostgreSQL for persistent workflow state and user management; and (3) FastAPI REST Endpoints connecting the Next.js frontend to the RAG and static audit engines.

5.5 Visual Illustration 3: Central Web Portal Review Workflow Wireframe

Illustration 3



Central web portal – review workflow and approval queue wireframe (Creator view on the left, Manager view on the right).

Creator Submission & Audit View	Manager Review & Approval View
Document: "Q4_Product_Launch_Release.pdf" | Status: [In Review – v2] | Privacy Mode: [Gemini 2.5 Flash Cloud / Local Ollama Toggle]	**Manager Queue:** 3 Pending Drafts | Active Filter: [Requires Sign-Off]
Highlighted Draft Editor: "Our #1 AI platform [FLAG 1: Unverified Claim] is super cheap [FLAG 2: Banned Term & Tone] and cuts costs by 99.9% [FLAG 3: Fact Contradiction]." + Uploaded Banner_v1.png [FLAG 4: Off-Brand Hex #FF0055 & WCAG Contrast 2.8:1 FAIL]	**Side-by-Side Diff Compare:** Original Creator Text vs. AI-Suggested Compliant Rewrite with highlighted changes and diff markers.
Selected Flag #3 Inspector: Issue: unsupported statistical claim ("99.9%"). Cited Source: Approved Fact Sheet ID #F-104 ("Verified 42% average cost reduction in 2026 study") & Brand Guide p.12, Sec 3.1. Suggested Rewrite: "...reduces operational costs by an average of 42%." AI Disclosure: Verified by Static Fact Worker #2 + Rewritten by Gemini 2.5 Flash.	**Collaboration Panel:** Inline comment threads between Creator and Manager; Revision History timeline (v1 → v2); Override Justification Log for intentional stylistic choices.
Creator Actions: [Accept Rewrite] [Modify Manually] [Override with Justification Comment] → Submit to Manager	**Manager Actions:** [Add Comment] [Request Changes] [Approve for Publication] → Logs telemetry to Governance Analytics

6. RESPONSIBLE-DESIGN CONSIDERATIONS: PRIVACY, SECURITY & HUMAN OVERSIGHT

1. Protecting Confidential Brand Materials

Pre-release product announcements and embargoed campaigns require strict data protection. First, our Delta-Only Architecture ensures that full confidential documents do not need to be sent to external LLMs – only isolated sentences flagged by local static workers are transmitted. Second, organisations can switch on **Confidential Local Mode** (Ollama / WebLLM) with one click, keeping 100% of embeddings, static checks, and LLM synthesis inside their own local hardware with zero external network calls.

2. Explicit Disclosure of AI & External Services (Rule 7 & Submission Integrity)

- **In-Product Disclosure:** Every audit card clearly displays which deterministic worker flagged the issue and which AI model (e.g., Google Gemini 2.5 Flash or Local Ollama) generated the suggested correction, along with the confidence score and source citation.
- **Development-Time AI & Tool Disclosure:** During the official competition build window, Team Xnihilo will use Google Gemini models and Antigravity available models/IDE tools to accelerate code generation, UI component scaffolding, and debugging. In strict compliance with our Human-in-the-Loop (HITL) engineering policy, no AI-generated code is blindly committed: every API route, RAG pipeline, static algorithm, and UI component is manually reviewed, integration-tested, and verified by the responsible team member.

3. Human Oversight & Non-Destructive Control

BrandGuard AI is designed as a collaborative co-pilot, never an automated censor. The platform never auto-overwrites creator text or blocks publication without human review. Creators can exercise **Override with Justification** when breaking a stylistic rule intentionally (e.g., a creative campaign slogan), passing their reasoning directly to the human Brand Manager for final judgment.

4. Security, Access Control & Accessibility

Role-Based Access Control (Creator, Reviewer, Manager) and Supabase Row-Level Security (RLS) isolate organisational brand vaults and audit logs, while our own web interface adheres to WCAG 2.1 AA contrast and keyboard navigation standards.

7. DEVELOPMENT PLAN, ROLE ALLOCATION & EXPECTED IMPACT

7.1 Team Xnihilo Role Allocation & Individual Accountability

Team Member	Designated Role	Specific Technical Responsibilities & HITL Testing Scope
Subhajit Das	Team Leader – RAG & AI Architecture	Leads overall system architecture, builds the Branched Multimodal RAG ingestion and retrieval pipeline (PyMuPDF + ChromaDB/Qdrant + Gemini Embeddings), engineers the Delta-Only Gemini Synthesizer prompts, and leads end-to-end Human-in-the-Loop integration testing.
Sambu Chowdhury	Core AI Engineer – RAG & Static Audit Pipeline	Builds the Approved Facts SQL/Hybrid verification engine and implements the 4 Parallel Static Pre-Check Workers (Regex/Aho-Corasick terminology scanner, NLP claim extractor, OpenCV/Pillow K-Means Hex Delta-E visual checker, EasyOCR banner scanner, and deterministic WCAG 2.1 contrast math).
Rana Tarafdar	Full-Stack Engineer – Frontend & FastAPI Backend Bridge	Develops the FastAPI asynchronous REST routes (/upload-guidelines, /audit-content, /approve-draft, /analytics), configures the Supabase PostgreSQL schema (RBAC roles, revision history snapshots, inline comments, and telemetry logs), and connects Frontend API state.
Mousumi Bag	UI/UX Engineer – Frontend Review Portal & Approval Workflow	Builds the Next.js Interactive Side-by-Side Diff Editor, Manager Approval Queue UI, revision history viewer, and Governance Insights Analytics dashboard charts.

7.2 Phased Development Plan (Strictly Within Official Build Window)

Phase	Focus	Details
Phase 0	**Pre-Build Discovery & Planning** Completed prior to build window per Rule 2	Conducted user surveys, manager interviews, problem validation, architecture design, and prepared this Round 1 Plan. Zero product source code was written prior to the official build window (Rules 1 & 4).
Phase 1	**Core Foundation & Branched RAG** Round 2 Build – Part A	Set up FastAPI + Next.js repository; implement Brand Vault ingestion (PDF chunking, Gemini embeddings, Approved Facts SQL table, and Hex/Logo rule storage) using Gemini & Antigravity coding assistance with continuous manual unit testing.
Phase 2	**4 Parallel Static Workers & Delta-Only Gemini Synthesis** Round 2 Build – Part B	Implement asyncio.gather parallel execution for the 4 static workers + Branched RAG lookup + single-pass Gemini 2.5 Flash synthesis returning structured JSON citations and rewrites.
Phase 3	**Role-Based Review Workflow & MVP Deployment** Round 2 Deliverable	Connect the Interactive Diff Editor, Accept/Modify/Override controls, Manager Approval sign-off, and revision history; deploy live judge-accessible MVP link and public Git repository.
Phase 4	**Stress Testing & Final Polish** Round 3 Finish	Deliver the complete OpenCV/WCAG visual accessibility suite, Confidential Local Mode toggle, and Governance Insights Dashboard; conduct structured user testing, log reliability improvements, and record the final demonstration.

7.3 Expected Organisational Impact

70%	**Faster Review Turnaround** Pre-screening drafts in ~2.5 seconds eliminates manual line-by-line manager fact-checking and hex-code inspection.
85%	**Lower AI Compute & Server Load** Filtering content through 4 deterministic static workers and sending only flagged deltas to a single Gemini call prevents server bottlenecks and rate-limit failures.
100%	**Citation Transparency & Compliance** Every flagged issue is backed by an exact page/section citation or mathematical WCAG/Hex proof, empowering creators while protecting brand trust.